Report

Of

DESIGN AND DEVELOPMENT OF A FULL-STACK WEB PLATFORM FOR AN ARCHITECTURE AND INTERIOR DESIGN FIRM

Six-Month Internship Report on the Full-Stack Developer & Design Intern role undertaken at Limitless Spacez, Greater Noida

Submitted To

School of Computer Science and Engineering
IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

Bachelor of Technology in Computer Science and Engineering

By

Name of Student: **SHOBHIT SINGH**
Roll Number of Student: 2410030993
Section: 3CSE19
Batch: 2024–28

2026–27

CANDIDATE’S DECLARATION

I, Shobhit Singh, do hereby declare that the internship report titled “Design and Development of a Full-Stack Web Platform for an Architecture and Interior Design Firm” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in Computer Science and Engineering. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree or diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

The internship was undertaken at Limitless Spacez, Greater Noida, Uttar Pradesh, in the capacity of Full-Stack Developer & Design Intern, for a period of six months with effect from 05 January 2026 to 04 July 2026, under the supervision of Ar. Sanjay Gautam, Founder & Principal Architect.

Signature: ______________________________

Student Name: Shobhit Singh
Roll No.: 2410030993
Section: 3CSE19
Date: ______________________

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the internship programme. I am thankful for their aspiring guidance, invaluably constructive criticism and friendly advice extended to me during the course of this work.

I would like to sincerely thank Ar. Sanjay Gautam, Founder & Principal Architect of Limitless Spacez, for offering me this internship, for reviewing my work at every stage, and for allowing me to take ownership of a live client-facing platform as a student. I am equally grateful to the design and site teams at Limitless Spacez, whose patience in explaining how an architecture practice actually works shaped almost every product decision described in this report.

I applied for this internship independently and carried out the work documented in this report on my own, and I am grateful to Limitless Spacez for placing that degree of trust in a student.

Finally, I express my heartfelt gratitude to my parents for their continued blessings, encouragement and support.

Signature: ______________________________

Student Name: Shobhit Singh
Roll No.: 2410030993

INTERNSHIP COMPLETION CERTIFICATE

LIMITLESS SPACEZ

DESIGN WITHOUT BOUNDARIES, BUILD WITHOUT LIMITS

Office No. UGF-50, Omaxe Arcade, Sector Omega-I, Greater Noida, G.B. Nagar - 201310 · +91 70489 04813 · sgaut285@gmail.com

Ref. No. LS/HR/CERT/2026/06

Date: 04 July 2026

CERTIFICATE OF INTERNSHIP COMPLETION

This is to certify that

Shobhit Singh

served Limitless Spacez, Greater Noida, as **Full-Stack Developer & Design Intern** for a period of six months, from **05 January 2026** to **04 July 2026**. During this engagement, the intern conceived, designed and developed the firm's public web platform at limitlessspacez.com, together with its administration panel and supporting documentation, and is commended for the technical rigour, professional discipline and design judgement demonstrated throughout.

LIMITLESS SPACEZ

Gautam

Proprietor

Ar. Sanjay Gautam

Founder & Principal Architect · COA Reg. No. CA/2021/140523

Limitless Spacez, Greater Noida, Uttar Pradesh

TABLE OF CONTENTS

4. Project Description

4.1 Introduction

An architecture and interior design practice sells work that clients cannot inspect in advance. A prospective homeowner cannot walk through a house that has not been built, and cannot judge a firm on a price list. The decision is made almost entirely on evidence of past work and on the confidence the practice inspires. For a small or mid-sized firm, the website is where that evidence is presented, and it is very often the only version of the practice a client sees before the first meeting.

This report documents a six-month internship undertaken at Limitless Spacez, an architecture, interior design and construction firm based in Greater Noida, Uttar Pradesh, between 05 January 2026 and 04 July 2026. The internship was offered in the capacity of Full-Stack Developer & Design Intern on a full-time hybrid basis, reporting to Ar. Sanjay Gautam, Founder & Principal Architect. The engagement was not a training exercise on a sample problem. The deliverable was the firm's live public platform at limitlessspacez.com, together with the internal tooling required for the firm to keep that platform current without a developer present.

The problem the internship set out to solve had two halves, and both had to be solved by the same person. The first half was a design problem: how a practice whose value lies in spatial judgement should be represented on a screen, so that photographs of completed projects are the subject of the page rather than decoration around text. The second half was an engineering problem: how to build a platform where every project, photograph and enquiry is data rather than hand-edited markup, so that adding a completed project is an ordinary office task instead of a code change.

The work was organised into four phases across the six months, each phase producing something the firm could use immediately. Phase one established the design system and the information architecture of the site. Phase two implemented the public site across its four service verticals. Phase three added the server, the database and the enquiry pipeline that turns a visitor into a recorded lead. Phase four delivered the content management panel, performance and search optimisation, and the handover documentation.

Working inside a practising design studio rather than a software team changed how the work had to be justified. Requirements did not arrive as tickets. They arrived as observations from an architect about how a client had misread a drawing, or from the site team about which photographs a client always asks to see. Translating those observations into a specification, and then defending a technical decision to people who judge proportion and material for a living, was as much a part of the internship as the code itself.

This report consolidates the six months into a single account. It describes the organisation, states the problem and objectives, defines the scope, documents the technology stack and system architecture, sets out the phase-wise methodology and its findings, records the outcomes achieved against those originally expected, and closes with the skills gained, the limitations encountered, and the work that remains.

4.2 Organization Profile

Limitless Spacez is a premium architecture, interior design and construction firm operating out of Greater Noida, Uttar Pradesh. The practice presents itself under the line "Design · Build · Deliver", which is an accurate description of its commercial position: it takes a project from initial design through statutory approval to completed construction, rather than handing the client off to separate agencies at each stage. The firm is led by Ar. Sanjay Gautam, Founder & Principal Architect, registered with the Council of Architecture under registration number CA/2021/140523.

The practice organises its offering into four service verticals, and this structure became the organising principle of the entire platform:

- **Architecture** — residential, commercial and institutional design from concept through working drawings.
- **Interior** — interior design, material and furniture specification, and turnkey fit-out.
- **Map Approval** — preparation and submission of building plans for sanction by the relevant local authority.
- **Construction** — execution and site supervision through to handover.

The Map Approval vertical is worth noting because it is unusual on an architecture firm's website and it materially affected the design. It is a compliance service, not an aesthetic one. Clients who arrive looking for it are anxious rather than inspired, and they are searching for a specific procedural answer. That vertical therefore needed a different page structure from the other three: process steps, document checklists and timelines rather than photography.

The firm's existing digital presence at the start of the internship consisted of an Instagram account used as the de facto portfolio, and a small static site. Instagram was doing real commercial work for the practice, and the platform was therefore designed to complement it rather than replace it, with the feed presented as current activity alongside a permanent, structured project archive on the site itself.

Organisationally the firm is small, which shaped every decision about maintainability. There is no in-house IT function and no retained developer. Any part of the platform that could only be updated by writing code would, in practice, stop being updated the day the internship ended. This constraint, more than any technical preference, is the reason the project was built as a database-backed application with an administration panel rather than as a larger set of static pages.

4.3 Problem Statement

A design practice that cannot show its work convincingly, cannot be found by the clients searching for it, and cannot update its own website, loses enquiries it has already earned. At the start of the internship the firm faced four connected problems.

Presentation. The practice's work is photographic and spatial, but the existing site presented it as small images embedded in blocks of text. Photographs were compressed into thumbnails, and there was no consistent way to present a project as a coherent narrative from brief to completion. The site described the firm; it did not demonstrate it.

Enquiry capture. Enquiries reached the firm through Instagram direct messages and phone calls. Nothing was recorded in a structured form, so the firm could not say how many enquiries it received in a month, which service they were for, or which of them had gone unanswered. A lead that arrives and is forgotten is indistinguishable from a lead that never arrived.

Maintainability. Every content change required editing markup. In a practice with no developer, this meant that completed projects, the strongest sales material the firm has, accumulated on a phone and never reached the website.

Discoverability and performance. The site carried large unoptimised images and minimal structured metadata. Clients in Greater Noida searching for an architect or for map approval assistance were unlikely to find the practice, and those who did arrive on a mobile connection waited for pages that were heavier than they needed to be.

The internship addressed these four problems as one problem: the firm needed a platform in which its work is structured data, presented through a design system built for photography, served quickly, indexed properly, and editable by the people who own it.

4.4 Project Objectives

- To study the firm's four service verticals and its enquiry process, and to derive from them an information architecture that a first-time visitor can navigate without instruction.
- To design a visual and typographic system in Figma in which photography is the primary content and interface elements are deliberately quiet, and to document it as reusable tokens and components.
- To implement a responsive front end for the public platform using semantic HTML5, CSS3 and Tailwind CSS, functional from a 360-pixel mobile viewport upward.
- To design a normalised relational schema in MySQL representing services, projects, project images and enquiries, and to build a Node.js and Express application over it.
- To implement a validated enquiry pipeline that records every submission in the database, notifies the firm by email, and defends against spam and injection.

- To build an authenticated administration panel through which non-technical staff can create and edit projects, upload and reorder images, and review enquiries, so that the platform can be maintained after the internship ends.
- To optimise delivery through responsive image handling, modern image formats, lazy loading and caching, targeting measurable improvement in Core Web Vitals on mobile.
- To implement on-page and technical search optimisation appropriate to a locally focused practice, including per-page metadata, semantic heading structure, structured data and a generated sitemap.
- To apply accessibility practice consistent with WCAG 2.1 Level AA in colour contrast, keyboard operability, focus visibility and alternative text.
- To follow a disciplined version-control and documentation practice using Git and GitHub, and to hand over a deployed platform with written operating documentation for the firm.

4.5 Scope of the Project

The scope of the internship covered the complete public web platform of Limitless Spacez and the internal tooling directly supporting it. It is summarised below in terms of what was included and, equally importantly, what was deliberately excluded.

Included in scope

- **Design.** A complete design system in Figma: type scale, colour, spacing, grid, component library, and high-fidelity layouts for every page template at mobile, tablet and desktop breakpoints.
- **Public front end.** Home page, the four service vertical pages, a filterable project archive, an individual project page template, an about page, and a contact page.
- **Server and data layer.** An Express application serving templated pages from MySQL, with a normalised schema, migration scripts and seed data.

- **Enquiry pipeline.** A validated contact and enquiry form with server-side validation, spam mitigation, database persistence and email notification to the firm.
- **Administration panel.** Session-based authenticated access for staff, with create, edit and delete operations on projects, image upload with ordering and captioning, and a read-and-annotate view of enquiries.
- **Performance, search and accessibility.** Image pipeline, caching, metadata, structured data, sitemap and robots configuration, and accessibility remediation.
- **Deployment and handover.** Production deployment, domain and certificate configuration, analytics instrumentation, backup procedure, and written documentation for the firm.

Excluded from scope

- Online payments, client billing and any financial transaction handling.
- A client-facing project tracking portal with per-client logins; this was discussed and deferred, and appears in Section 4.13.
- Native mobile applications; the platform is a responsive web application only.
- Architectural drawing production, BIM or CAD work, and any professional service the firm itself delivers.
- Migration of the firm's internal accounting or project management records.

One further boundary should be stated plainly. All photography, project descriptions and firm credentials published through the platform were supplied and approved by Limitless Spacez. No client name, drawing or site detail was published without the firm's explicit approval, and no confidential client material appears in this report.

4.6 Technologies and Tools Used

The stack was chosen against one dominant constraint: the platform had to remain operable by a small firm with no developer on staff after the internship ended. That ruled out anything requiring a build pipeline the firm could not run or a hosting model it could not afford to keep current. A server-rendered Express application over MySQL, with Tailwind compiled to a single stylesheet, was the smallest arrangement that met every objective while remaining conventional enough to be picked up by any future developer. Table 4.1 lists the technologies actually used and the purpose each served.

Layer	Technology / Tool	Purpose in this project
Design	Figma	Design system, component library, responsive layouts for every template, developer handoff specifications
Markup & styling	HTML5, CSS3	Semantic document structure, CSS Grid and Flexbox layout, custom properties for design tokens
Utility framework	Tailwind CSS	Design tokens expressed as configuration, consistent spacing and type scale, purged production stylesheet
Client scripting	JavaScript (ES2022)	Navigation, project filtering, lightbox gallery, form validation feedback, lazy-loading behaviour
Server runtime	Node.js	Application runtime for the public site, the enquiry pipeline and the administration panel
Web framework	Express	Routing, middleware, server-side templating, session handling, file upload endpoints
Database	MySQL	Normalised storage of services, projects, project images, enquiries and staff accounts
Version control	Git, GitHub	Branch-per-feature workflow, review history, issue tracking, deployment source of truth
Hosting & analytics	Managed Node hosting, TLS, Google Analytics 4, Search Console	Production deployment, certificate management, traffic and enquiry-conversion measurement, index monitoring

Table 4.1. Technologies and tools used across the internship, by layer.

Two choices in Table 4.1 were argued over and are worth justifying. Tailwind CSS was adopted not for the speed of writing utility classes but because its configuration file is a written record of the design system: the type scale, spacing rhythm and palette defined in Figma exist in exactly one place in the codebase, which removed the usual drift between design file and implementation. MySQL was chosen over a document database because the data is genuinely relational, a project belongs to a service and owns an ordered set of images, and because managed MySQL is available on every hosting plan the firm would plausibly ever use.

No paid third-party service was introduced that the firm would have to renew, other than hosting and the domain it already held. This was a deliberate constraint: every recurring cost added during an internship is a cost the firm must justify without the intern present to explain it.

4.7 System Architecture

The platform follows a conventional three-tier server-rendered architecture. A visitor's request reaches the Express application, which resolves the route, queries MySQL for the content that route needs, renders a template on the server, and returns HTML. Static assets and processed images are served directly, with caching headers set by asset type. Figure 4.1 shows the arrangement of layers.

Server-side rendering was chosen rather than a client-rendered single-page application for two reasons specific to this project. First, the pages are content pages whose primary job is to be indexed and to display photographs quickly on a mobile connection, which server-rendered HTML does better with less machinery. Second, a server-rendered application has far fewer moving parts for a future maintainer to understand, which was a stated objective rather than an afterthought.

Client tier — browser

Responsive HTML delivered from the server · Tailwind stylesheet · progressive JavaScript for navigation, project filtering, lightbox gallery and inline form validation · native lazy loading for below-fold imagery

↓ HTTPS request / rendered HTML response ↓

Application tier — Node.js and Express

Public routes: home, four service verticals, project archive, project detail, about, contact

Middleware: security headers, request logging, body parsing, session management, rate limiting

Enquiry pipeline: server-side validation → persistence → email notification to the firm

Admin routes (authenticated): project CRUD, image upload and ordering, enquiry review

↓ parameterised queries ↓

Data tier — MySQL

services · projects · project_images · enquiries · users | scheduled backup of schema and data

↓ served directly with cache headers ↓

Asset layer

Processed project photography in multiple widths and modern formats · fonts · compiled stylesheet · client scripts

Figure 4.1. Layered architecture of the Limitless Spacez web platform.

The data model is deliberately small. A `services` table holds the four verticals. A `projects` table holds one row per completed or ongoing project, with a foreign key to its service, a URL slug, location, year, status, a short summary and a longer description. A `project_images` table holds one row per photograph with a foreign key to its project, an explicit sort order, a caption and alternative text. An `enquiries` table records every form submission with the service enquired about, the visitor's message and contact details, a timestamp, and a status field the firm updates as it follows up. A `users` table holds staff accounts with hashed passwords.

Two details of that model came directly from working alongside the design team. The explicit sort order on images exists because the sequence in which a project is shown is an editorial decision an architect wants to control, not an accident of upload time. The separate alternative text field exists because a caption written for a client and a description written for a screen reader are different pieces of writing, and collapsing them produces bad versions of both.

Security is handled at the layer where it belongs rather than at the form. Every database access uses parameterised queries, so user input is never concatenated into SQL. Output is escaped at render time. Passwords are stored only as salted hashes. Sessions use signed, HTTP-only cookies. Upload endpoints validate file type and size and rewrite images through the processing pipeline rather than storing what was uploaded. Security headers, including a content security policy, are set as middleware so they apply to every route by default rather than being remembered per page.

4.8 Methodology

The internship followed an incremental, phase-wise method rather than a single design-then-build sequence. Each of the four phases ended in something the firm could see and use, and each phase's review with the Principal Architect set the priorities of the next. This mattered because the client was also the supervisor: feedback arrived continuously, and a method that deferred all review to the end would have wasted months. Table 4.2 summarises the phases; the sub-sections that follow describe the work and findings of each.

Phase	Period	Focus	Deliverable at review
1	Jan – Feb 2026	Discovery, information architecture, design system and high-fidelity layouts	Approved Figma design system and page designs at three breakpoints
2	Feb – Mar 2026	Front-end implementation of all public templates, responsive and accessible	Working static front end reviewed on real devices
3	Apr – May 2026	Express application, MySQL schema, dynamic content and the enquiry pipeline	Database-driven site with recorded enquiries and email notification
4	May – Jul 2026	Administration panel, performance and search optimisation, deployment, handover	Live platform, staff training session and written documentation

Table 4.2. Phase-wise plan of the six-month internship and the deliverable reviewed at the end of each phase.

4.8.1 Phase 1 — Discovery, Information Architecture and Design System

Phase one began with structured conversations rather than sketches. I sat with the Principal Architect and the site team to establish what a client actually asks in a first meeting, which questions recur, and what evidence closes an enquiry. Three findings from that discovery shaped everything afterwards.

First, clients do not browse by service; they browse by resemblance. A client planning a house wants to see houses, and only then asks who did the interiors. This argued against a navigation built purely around the firm's four verticals and in favour of a project archive filterable by both type and service, with the vertical pages acting as explanations of process rather than as galleries.

Second, the Map Approval enquiry is a different conversation entirely. Those clients want procedure, documents and timelines. That page was designed as a process page with a clear document checklist and a direct enquiry form, not as a portfolio page.

Third, almost all traffic would be mobile, arriving from Instagram. Designing desktop-first and scaling down would have been the wrong order, so every layout was designed at a 360-pixel viewport first and expanded upward.

The information architecture that followed was intentionally shallow: no page of consequence sits more than two clicks from the home page. The design system was then built in Figma as a token set and component library. Its governing rule was restraint. Photography carries the practice, so the interface around it uses a near-monochrome palette, generous whitespace, a single typographic pairing with a strict scale, and a twelve-column grid whose gutters were sized to let images sit edge to edge without competing borders.

Every colour pair in the system was checked for contrast against WCAG 2.1 Level AA at the point of definition rather than after implementation, which removed an entire class of rework later. Components were built with defined states, including focus states, so that keyboard operability was a property of the design system rather than something retrofitted in code.

Phase one closed with a review at which the design system and the full set of page layouts were approved with two changes: the home page hero was simplified to a single full-bleed photograph, and project cards lost their captions in favour of a hover and focus reveal, on the argument that a grid of photographs reads faster than a grid of photographs plus text.

4.8.2 Phase 2 — Front-End Implementation

Phase two implemented every approved template as working front end. The design tokens defined in Figma were transferred into the Tailwind configuration first, so that no spacing value, type size or colour could enter the codebase without existing in the system. Layout was built with CSS Grid and Flexbox, and markup was written semantically: a single `h1` per page, sectioning elements that describe the document rather than its appearance, and landmark regions for navigation, main content and footer.

JavaScript was written as an enhancement rather than a dependency. The navigation, project filter, lightbox and form feedback all improve the experience, but every page remains readable and every form remains submittable with scripting disabled. This mattered practically as well as principally: search crawlers and slow connections see the same content the visitor does.

Testing in this phase was done on real devices, not only in the browser’s responsive simulator, and this produced the phase’s most useful correction. On a mid-range Android handset held one-handed, the project filter controls sat within reach but the enquiry call to action did not.

The layout was revised so that the primary action on every mobile template falls inside the comfortable thumb zone, and all interactive targets were audited to a minimum of forty-four pixels. A second correction came from testing with the keyboard alone: the lightbox trapped focus incorrectly and could not be dismissed without a mouse. Fixing it properly, with focus containment while open and focus restoration on close, took longer than building the lightbox had.

Phase two ended with the complete public front end working from static placeholder content, reviewed on desktop, tablet and two handsets.

4.8.3 Phase 3 — Server, Database and Enquiry Pipeline

Phase three replaced placeholder content with real data. The MySQL schema described in Section 4.7 was designed on paper first, normalised to third normal form, then created through migration scripts held in version control so that the database could be rebuilt from scratch reproducibly. Seed data was written for the four services and for the first set of real projects supplied by the firm.

The Express application was then built over that schema. Routes were kept thin: a route resolves parameters, calls a query function, and renders a template. Query functions were isolated in a data-access module so that no SQL is written inside a route handler, which made both testing and later review straightforward. Templates were server-rendered from the same markup produced in phase two, which meant the transition from static to dynamic changed almost nothing visible.

The enquiry pipeline received disproportionate attention because it is the only part of the platform whose failure costs the firm money directly. Client-side validation exists for immediacy, but every field is validated again on the server, because client-side validation is a convenience and not a control.

A valid submission is written to the `enquiries` table inside a transaction and only then triggers an email notification to the firm, so that a mail delivery failure can never cause a lost lead. Rate limiting per address and a hidden honeypot field handle automated spam without imposing a challenge on genuine visitors. The visitor sees a confirmation that states plainly what will happen next and by when, which was the Principal Architect's request rather than mine, and a good one.

Phase three closed with the platform serving all content from the database and every enquiry recorded and notified. The firm could, for the first time, answer the question of how many enquiries it had received in a week and for which service.

4.8.4 Phase 4 — Administration Panel, Optimisation and Handover

Phase four built the part of the project that determines whether any of the rest survives. The administration panel is a session-authenticated area in which staff can add a project, write its description, upload photographs, drag them into the intended order, write captions and alternative text, publish or unpublish, and read enquiries while marking each as new, contacted or closed.

The panel was designed under a rule borrowed from the discovery interviews: anything that requires an explanation will not be used. Field labels use the vocabulary of the practice rather than of the database, destructive actions require confirmation and are recoverable, and image upload handles the multi-select of photographs straight from a phone, because that is how the photographs actually arrive. The panel was tested by watching a member of staff add a real project unaided, and the two places where they hesitated were redesigned.

Optimisation followed. Uploaded photography is processed on upload into several widths in a modern format with a compatible fallback, and served through responsive image markup so that a phone never downloads a desktop-sized file.

Below-fold images are lazily loaded, dimensions are declared on every image to eliminate layout shift, the stylesheet is purged of unused utilities before deployment, fonts are subset and preloaded, and static assets are served with long-lived cache headers. Each change was measured rather than assumed, using synthetic auditing on a throttled mobile profile, and any change that did not measurably help was reverted.

Search optimisation was treated as a structural task, not a keyword task. Every page received a purpose-written title and description, a single coherent heading hierarchy, and canonical URLs. Structured data was added describing the practice as a local business with its services and location, since a firm serving Greater Noida is found through local search far more than through generic terms. A sitemap is generated from the database, so a newly published project becomes discoverable without anyone remembering to update a file, and the property was verified in Search Console so that indexing could be monitored rather than guessed at.

The phase, and the internship, ended with deployment and handover: production deployment with TLS on the firm's domain, analytics configured with the enquiry submission recorded as a conversion event, a scheduled database and image backup with a documented restore procedure, and written documentation covering how to add a project, how to interpret the enquiry list, how the backup works and what to do if the site is unreachable. A training session was held with the staff who would use the panel, and the documentation was revised afterwards to answer the questions they actually asked.

4.8.5 Working Practice

Throughout all four phases the work followed a consistent practice: a branch per feature, self-review before every merge, descriptive commit history, and issues tracked in the repository so that the firm has a written record of why each decision was taken.

4.9 Expected and Achieved Outcomes

The outcomes expected at the start of the internship, and the position at its conclusion, are set out below. Where an expected outcome was not fully met, this is stated rather than reframed.

A live, database-driven platform. Achieved. The public platform is deployed on the firm’s domain over TLS, serving all content from MySQL through the Express application, across the home page, four service verticals, a filterable project archive, individual project pages, an about page and a contact page.

A documented design system. Achieved. The Figma library of tokens and components exists as a maintained artefact, and its tokens are the Tailwind configuration rather than a separate parallel definition, which means the design system and the implementation cannot drift apart silently.

A structured enquiry pipeline. Achieved. Every submission is validated on the server, persisted, notified by email and tracked through a status field. The firm now has a record of enquiry volume by service, which it did not have before, and the beginning of an evidence base for where its marketing effort is worth spending.

Independent content maintenance. Achieved. Staff add and edit projects, order and caption images, publish and unpublish, and review enquiries through the administration panel without developer involvement. This was demonstrated in the handover session, and it is the outcome I would defend as the most valuable of the six months.

Measurable performance improvement on mobile. Achieved. Responsive image handling, modern formats, lazy loading, declared dimensions, stylesheet purging and caching produced substantial measured improvement against the starting baseline on a throttled mobile profile, with layout shift effectively eliminated.

Technical search foundation. Achieved as a foundation, not as a result. Metadata, heading structure, canonical URLs, local-business structured data, a database-generated sitemap and Search Console verification are all in place. Ranking, however, accrues over months of published work and cannot be claimed as an outcome of a six-month internship; what exists is the condition for it.

Accessibility to WCAG 2.1 Level AA. Substantially achieved. Contrast, keyboard operability, focus visibility, landmark structure and form labelling meet the standard, and automated auditing reports no violations on any template. Full conformance nevertheless depends on content authored after handover, particularly on alternative text being written for every uploaded photograph. The panel prompts for it, but it cannot enforce quality, and this is recorded honestly as a limitation in Section 4.12.

Handover documentation and continuity. Achieved. The repository, migration scripts, backup and restore procedure, and written operating documentation were handed to the firm, with a training session for the staff who use the panel.

4.10 Certificates and Supporting Proof

The following documents evidence the internship described in this report and are attached to it:

- Letter of Offer, reference LS/HR/INT/2026/06 dated 03 January 2026, issued by Limitless Spacez and signed by Ar. Sanjay Gautam, Founder & Principal Architect, offering the position of Full-Stack Developer & Design Intern with effect from 05 January 2026 for six months.
- Internship Completion Certificate issued by Limitless Spacez on conclusion of the engagement, attached in the front matter of this report.

4.11 Skills Acquired

Table 4.3 records the skills developed during the internship and, in each case, the work in which they were exercised.

Area	Skill	Where it was exercised
Product & UX	Discovery interviewing, information architecture, mobile-first layout	Phase 1 discovery with the architect and site team; navigation and archive structure
Visual design	Design systems, type and spacing scales, component libraries in Figma	Token set and component library carried through to the Tailwind configuration
Front end	Semantic HTML5, CSS Grid and Flexbox, Tailwind, progressive enhancement	All public templates; scripting written as enhancement rather than dependency
Back end	Node.js and Express, routing and middleware, server-side templating, sessions	Public routes, enquiry pipeline, authenticated administration panel
Data	Relational modelling, normalisation, parameterised queries, migrations	MySQL schema, reproducible migration scripts, data-access module
Security	Input validation, injection and XSS defence, password hashing, upload validation, security headers	Enquiry pipeline, authentication, image upload endpoints, global middleware
Performance	Responsive images, modern formats, lazy loading, caching, measurement discipline	Phase 4 optimisation, measured on a throttled mobile profile before and after
Search & analytics	On-page and technical SEO, structured data, analytics and conversion events	Metadata, local-business schema, generated sitemap, Search Console, GA4
Accessibility	WCAG 2.1 AA practice, keyboard operability, focus management	Contrast decided in the design system; lightbox focus containment rebuilt
Engineering practice	Git branching, self-review, issue tracking, deployment, backup and restore	Full six months; production deployment and documented handover
Professional	Client communication, scope negotiation, technical writing for non-technical readers	Phase reviews with the Principal Architect; operating documentation and training

Table 4.3. Skills acquired during the internship and the work in which each was exercised.

4.12 Challenges and Limitations

Working to a design-led client. The most instructive difficulty of the internship was that my supervisor evaluated the work by eye, and by a trained eye. Feedback often arrived as a judgement rather than a specification: a layout felt crowded, a photograph did not have enough air around it, a typeface felt commercial. Learning to convert that into a change in a spacing token or a type scale, rather than into an ad hoc adjustment on one page, took most of phase one. It is also the skill I value most from the six months.

Photography quality and consistency. A portfolio platform is only as good as its images. The available photography varied widely in resolution, aspect ratio, white balance and quality, some of it originating from phones on site. Since reshooting completed projects was not possible, the design had to accommodate this: the image pipeline normalises dimensions and the layouts were designed to tolerate inconsistent crops without appearing broken. This is mitigation, not a solution, and consistent project photography remains the single highest-value improvement available to the firm.

Content is the bottleneck. Building the project archive was fast; filling it was not. Project descriptions require time from senior staff who are billing on live projects, and that time is genuinely scarce. The archive launched with fewer projects than it was designed to hold. In response, the templates were revised to look considered rather than empty at low volume, and the panel was built to make adding one project a task of minutes.

Working alone on a full stack. Being the only person on design, front end, back end and deployment meant there was no reviewer to catch my mistakes. I compensated with a branch-per-feature workflow, deliberate self-review against a written checklist before each merge, and testing on real devices, but the absence of a second technical opinion is a real limitation of a single-intern project and I would not claim otherwise.

Accessibility depends on future authoring. As noted in Section 4.9, alternative text is authored per photograph after handover. The panel requires the field and explains what belongs in it, but it cannot judge whether what is written is useful. Sustained conformance is therefore an operational commitment by the firm, not a property of the code.

Search results take longer than an internship. The technical foundation for local search is complete, but visibility accrues over months of published work and accumulated local signals. The internship can be judged on what was built; it cannot fairly be judged on ranking achieved within its own duration.

Scope pressure. Several attractive features were proposed mid-project, including a client project-tracking portal and a blog. Agreeing to defer them, rather than accepting them and delivering everything half-finished, was an uncomfortable but correct decision, and they are recorded below as future scope rather than as failures.

4.13 Future Scope

The platform was built to be extended, and the following work is recommended in order of value to the practice.

A client project portal. The clearest next step, and the feature the firm asked about most. A per-client authenticated area showing stage of work, approved drawings, site photographs and payment milestones would reduce the volume of status telephone calls the studio handles.

The existing schema and authentication already provide most of the foundation for it, and it was deferred for time rather than for feasibility.

A map approval process tracker. Because the Map Approval vertical is procedural, a simple public tracker showing the stage a submission has reached, together with the documents required at each stage, would answer the most frequent question those clients ask and would differentiate the practice from competitors offering the same service.

Enquiry analytics on the existing data. The `enquiries` table now accumulates structured data that nothing yet reads. A modest dashboard reporting volume by service, by month and by source, and time to first response, would let the firm direct its marketing spend on evidence. This is the cheapest of all the recommendations and probably the most immediately useful.

Editorial content. A written section covering completed project case studies and guidance on topics such as the approval process would serve both prospective clients and local search, using content the practice already possesses as expertise. It requires an editorial commitment from the firm before it requires any code.

A structured photography standard. A short shooting brief for completed projects, covering required shots, orientation and lighting conditions, would remove the inconsistency described in Section 4.12 at source rather than compensating for it in software.

Operational hardening. Two further measures are recommended: automated verification that the scheduled backup restores successfully, rather than merely that it ran, and continuous uptime and performance monitoring so that a regression is reported by the system rather than noticed by a client.

4.14 Conclusion

The six-month internship at Limitless Spacez delivered a live, database-driven web platform for a practising architecture and interior design firm, together with the design system it is built on, the enquiry pipeline that captures its leads, the administration panel through which its staff maintain it, and the documentation that allows all of this to continue without me.

Measured against the objectives stated in Section 4.4, the technical outcomes were met. The platform is deployed and serving content from MySQL through an Express application; the front end is responsive, semantic and substantially accessible; enquiries are validated, recorded and notified; performance on mobile improved measurably against a documented baseline; and the technical foundation for local search discoverability is complete. Two objectives are honestly reported as partially met, both for reasons outside the code: full accessibility conformance depends on content authored after handover, and search visibility accrues over a period longer than the internship itself.

The more durable lessons were not technical. The first is that a platform which cannot be maintained by the people who own it will decay regardless of how well it is engineered, which is why the administration panel and the handover documentation, the least visible parts of the project, are the parts I would defend first. The second is that requirements in a real practice do not arrive as specifications; they arrive as a professional's judgement about their own work, and an engineer's job is to translate that judgement into structure without flattening it. The third is that constraints, in this case a small firm with no developer, no budget for recurring services and photography that could not be reshot, are not obstacles to a design but the material from which the design is made.

The internship was the first occasion on which my work was used by people who had not asked how it was built and did not need to care. That changes the standard, and applying that standard is the most useful thing I take from these six months into the remainder of my degree.

5. Bibliography / References

1. Limitless Spacez, *Limitless Spacez — Premium Architecture, Interior Design and Construction*. [Online]. Available: https://limitlessspacez.com/. [Accessed: 30 Jun. 2026].

2. Limitless Spacez, *Letter of Offer — Internship, Ref. No. LS/HR/INT/2026/06*, issued 03 January 2026, Greater Noida, Uttar Pradesh.

3. Mozilla Developer Network, *MDN Web Docs — HTML, CSS and JavaScript Reference*. [Online]. Available: https://developer.mozilla.org/. [Accessed: 22 Jun. 2026].

4. World Wide Web Consortium, *HTML Living Standard*, WHATWG. [Online]. Available: https://html.spec.whatwg.org/. [Accessed: 14 Feb. 2026].

5. OpenJS Foundation, *Node.js Documentation*. [Online]. Available: https://nodejs.org/docs/. [Accessed: 12 Apr. 2026].

6. OpenJS Foundation, *Express — Node.js Web Application Framework Guide*. [Online]. Available: https://expressjs.com/. [Accessed: 12 Apr. 2026].

7. Oracle Corporation, *MySQL 8.0 Reference Manual*. [Online]. Available: https://dev.mysql.com/doc/refman/8.0/en/. [Accessed: 19 Apr. 2026].

8. Tailwind Labs, *Tailwind CSS Documentation*. [Online]. Available: https://tailwindcss.com/docs. [Accessed: 03 Mar. 2026].

9. Google, *web.dev — Core Web Vitals and Performance Guidance*. [Online]. Available: https://web.dev/. [Accessed: 08 Jun. 2026].

10. Google, *Google Search Central Documentation — SEO Starter Guide and Structured Data*. [Online]. Available: https://developers.google.com/search/docs. [Accessed: 15 Jun. 2026].

11. World Wide Web Consortium, *Web Content Accessibility Guidelines (WCAG) 2.1*, W3C Recommendation, 05 June 2018. [Online]. Available: https://www.w3.org/TR/WCAG21/. [Accessed: 27 Feb. 2026].

12. OWASP Foundation, *OWASP Top Ten Web Application Security Risks*. [Online]. Available: https://owasp.org/www-project-top-ten/. [Accessed: 30 Apr. 2026].

13. S. Chacon and B. Straub, *Pro Git*, 2nd ed. New York: Apress, 2014. [Online]. Available: https://git-scm.com/book/en/v2.

14. Figma Inc., *Figma Help Center — Design Systems, Variables and Developer Handoff*. [Online]. Available: https://help.figma.com/. [Accessed: 09 Feb. 2026].

ANNEXURE

Letter of Offer — Ref. No. LS/HR/INT/2026/06

LIMITLESS SPACEZ

DESIGN WITHOUT BOUNDARIES, BUILD WITHOUT LIMITS

Office No. UGF-50, Omaxe Arcade, Sector Omega-I, Greater Noida, G.B. Nagar – 201310 · +91 70489 04813 · [illegible]

Ref. No. LS/HR/INT/2026/06

Date: 03 January 2026

To,
Mr. Shobhit Singh
1203, Tower 3, White House Apartments
Greater Noida, Uttar Pradesh – 201310

SUBJECT: OFFER OF INTERNSHIP — FULL-STACK DEVELOPER & DESIGN INTERN

Dear Mr. Shobhit Singh,

With reference to your application and the subsequent interview, we are pleased to extend to you an offer of internship with Limitless Spacez in the capacity of **Full-Stack Developer & Design Intern**. The internship shall commence on **05 January 2026** and continue for a period of six months, concluding on **04 July 2026**, based at our office in Greater Noida, Uttar Pradesh.

This shall be a full-time engagement on a hybrid working basis, and you shall report directly to Ar. Sanjay Gautam, Founder & Principal Architect. In consideration of your services, you shall be paid a monthly stipend of **₹12,000**, payable in arrears on or before the 7th day of the following month. This offer is extended subject to the terms and conditions set out below.

LIMITLESS SPACEZ

Gautam

FOR LIMITLESS S Proprietor

SANJAY GAUTAM
Founder & Principal Architect
COA Reg. No. CA/2021/140523

Place: Greater Noida
Date: 05/Jan/2026

ACCEPTED BY THE INTERN

Shobhit Singh

SHOBHIT SINGH
Full-Stack Developer & Design Intern

Place: Greater Noida
Date: 05/Jan/2026